Exercise 2

On higher derivatives of the function

From the school book | Understand | Apply | Higher Order Thinking Skills

1 **Find $\dot{y}$ for each of the following :**

(1) $y = x^3 - 5x^2 + 6x - 4$

(2) $y = (3x^2 - 4)(x + 3)$

(3) $y = x^2 (x + 2)^2$

(4) $y = \dfrac{x+2}{x-2}$

(5) $y = x + \sqrt{x+5}$

(6) $y = \sqrt[3]{(x-2)^2}$

(7) $y = \sin(5x + 1)$

(8) $y = 2x^2 + \cos 3x$

(9) $y = x \cos 2x$

(10) $y = 9 \tan \dfrac{x}{3}$

(11) $y = 2 \sin^2 x - 5 \cos^2 x$

(12) $y = x \sin^2 x$

2 **Find $\dfrac{d^2 y}{dx^2}$ for each of the following at the indicated values of x :**

(1) $y = \sin x - \cos x$ at $x = \dfrac{\pi}{4}$ « zero »

(2) $y = 2x \sin x \cos x$ at $x = \pi$ « 4 »

(3) $y = x \cos x - \sin x$ at $x = \dfrac{\pi}{2}$ « – 1 »

(4) $y = \dfrac{2x-1}{x+3}$ at $x = -2$ « – 14 »

3 **Find the third derivative for each of the following :**

(1) $y = x^5 - 4x^3 + 3$

(2) $y = \sin x$

(3) $y = \cos(\pi - 3x)$

(4) $y = (2x + 7)^4 + \sin 2x$

(5) $f(x) = \dfrac{2x}{x+1}$

(6) $y = \dfrac{x+1}{x-1}$

(7) $y = 2x + \dfrac{3}{x}$

(8) $y = \sqrt{2x-5}$

(9) $y = \sin x \cos x$

(10) $y = 2x^3 + x \cos 2x$

4 If $y = x^5 - 10x^4 + 60x^2 + 120$, find value of $\dfrac{d^2 y}{dx^2}$ at $\dfrac{d^4 y}{dx^4}$ = zero « – 200 »

5 **Choose the correct answer from those given :**

(1) If $f(x) = x^3 - 5x^2 + 9x - 3$, then $f''(0) =$ .....

(a) – 20 (b) – 10 (c) 0 (d) 10

(2) *(2nd Session 2014)* If $f(x) = \sin 2x$, then $f''\left(\frac{\pi}{4}\right) =$ .....

(a) zero (b) – 2 (c) – 4 (d) – 6

(3) If $f(x) = a x^3 + 3x^2 + 4x + 1$ and $f''(1) = 6$, then a = ...

(a) zero (b) 2 (c) 4 (d) 6

(4) If $y = x^9 - 14x^7 - x^2 + 3$, then $\frac{d^{10}y}{dx^{10}} =$ .....

(a) zero (b) 9 (c) 18 (d) 72

(5) If $y = (5x - 4)(x + 3)$, $z = 3x^2 - 4x + 17$, then $\frac{d^2y}{dx^2} + \frac{d^2z}{dx^2} =$ .....

(a) 12 (b) 14 (c) 16 (d) 18

(6) $\frac{d^2}{dx^2}\left(\cos^2\frac{x}{2} - \sin^2\frac{x}{2}\right) =$ .....

(a) $-\cos x$ (b) $\cos x$ (c) $-\sin x$ (d) $\cos x - \sin x$

(7) *(Sudan 2014)* If $y = -\sin x$, then $\frac{d^2y}{dx^2} + y =$ .....

(a) – 4 (b) 2 (c) 4 (d) zero

(8) If $(x + y)^5 = 3$, then $\frac{d^2y}{dx^2} + \frac{dy}{dx} =$ .....

(a) – 1 (b) zero (c) 1 (d) $20(x + y)^4$

(9) *(1st Session 2024)* If $y = \frac{x}{x-3}$, then $\frac{d^3y}{dx^3} =$ ..... at $x = 4$

(a) – 18 (b) – 9 (c) 9 (d) 18

(10) $\frac{d}{dx}\left[y\frac{dy}{dx}\right] = y\frac{d^2y}{dx^2} +$ .....

(a) $\frac{d^2y}{dx^2}$ (b) $\left(\frac{dy}{dx}\right)^2$ (c) $\frac{dy}{dx}$ (d) y

(11) If f is a polynomial function of fifth degree , then the fifth derivative of the function f equals .....

(a) zero (b) non-zero constant (c) x (d) $5x$

(12) The degree of the third derivative of the quadratic function is a function .....

(a) of second degree. (b) of first degree.

(c) of zero degree. (d) with no degree.

(13) The second derivative of the cubic function is a ..... function.

(a) linear (b) quadratic (c) constant (d) zero

(14) If $S = a \cos(\ell t + m)$ is the equation of the motion of a particle where a , ℓ and m are constants , then $\frac{d^2 S}{d t^2}$ = ...

(a) ℓS (b) $\ell^2 S$ (c) $-\ell S$ (d) $-\ell^2 S$

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6 *(1st Session 2001)* If $x^2 + y^2 = 16$ **Prove that :** $y \frac{d^2 y}{d x^2} + \left(\frac{d y}{d x}\right)^2 + 1 = 0$

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7 *(2nd Session 2006)* If $y^2 + a x^3 - b x = c$ where a , b , c are constants

Prove that : $y \frac{d^2 y}{d x^2} + \left(\frac{d y}{d x}\right)^2 + 3 a x = 0$

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8 If $4 y^3 = 3 x^2$ **Prove that :** $2 y^2 \frac{d^2 y}{d x^2} + 4 y \left(\frac{d y}{d x}\right)^2 = 1$

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9 *(2nd Session 2002)* If $x^3 + y^3 = 1$ **Prove that :** $y^2 y'' + 2 y y'^2 + 2 x = 0$

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10 *(2nd Session 2012)* If $x^3 y^3 = 1$ **Prove that :** $x^2 \frac{d^2 y}{d x^2} = 2 y$

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11 *(2nd Session 2014)* If $x^2 y = 2 x + 5$ **Prove that :** $x^2 \frac{d^2 y}{d x^2} + 4 x \frac{d y}{d x} + 2 y = 0$

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12 *(2nd Session 2010)* If $y^2 = x$ **Prove that :** $x \frac{d^2 y}{d x^2} + y \left(\frac{d y}{d x}\right)^2 = 0$

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13 *(1st Session 2003)* If $2 x y + 3 = 5 x^2$ **Prove that :** $x \left(\frac{d^2 y}{d x^2}\right) + 2 \left(\frac{d y}{d x}\right) = 5$

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14 *(1st Session 2005)* If $y^2 = 1 - \frac{1}{x^2}$ **Prove that :** $y \frac{d^2 y}{d x^2} + \left(\frac{d y}{d x}\right)^2 + \frac{3}{x^4} = 0$

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15 If $y = \frac{1}{x}$, $x \neq 0$ **Prove that :** $x^2 \frac{d^2 y}{d x^2} + 3 x \frac{d y}{d x} + y =$ zero

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16 If $(1 + x) y = x^2$ **Prove that :** $(1 + x) \frac{d^2 y}{d x^2} + 2 \frac{d y}{d x} = 2$

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17 If $y^2 + x^2 y^2 = 8$ **Prove that :** $(x^2 + 1) \frac{d^2 y}{d x^2} + 3 x \frac{d y}{d x} + y = 0$

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18 *(2nd Session 2008)* If $4 x^2 + y^2 = 25$ **Prove that :** $y^3 \frac{d^2 y}{d x^2} + 100 = 0$

19 *(1st Session 2012)* If $x + y = x y$ **Prove that :** $\frac{d^2 y}{d x^2} = 2 \left(\frac{y}{x}\right)^3$

20 If $y = \sqrt{2 x + 5}$ **Prove that :** $(2 x + 5) \frac{d^3 y}{d x^3} + 3 \frac{d^2 y}{d x^2} = 0$

21 If $y \sqrt{1 + x^2} = 1$ **Prove that :** $(1 + x^2) \frac{d^2 y}{d x^2} + 3 x \frac{d y}{d x} + y = 0$

22 If $y^2 = \sqrt{4 x + 1}$, $x > \frac{-1}{4}$ **Prove that :** $3 \left(\frac{d y}{d x}\right)^2 + y \frac{d^2 y}{d x^2} = 0$

23 If $y^3 = 3 x^4 - 3 x^2 + 5$ **Prove that :** $y^2 \frac{d^2 y}{d x^2} + 2 y \left(\frac{d y}{d x}\right)^2 + 2 = 12 x^2$

24 *(2nd Session 2005)* If $y = \cos x$ **Prove that :** $\left(\frac{d y}{d x}\right)^2 - y \frac{d^2 y}{d x^2} = 1$

25 *(1st Session 2010)* If $y = 2 \cos (3 x + 1)$ **Prove that :** $\frac{d^2 y}{d x^2} + 9 y = 0$

26 If $y = \tan x$ **Prove that :** $\frac{d^2 y}{d x^2} = 2 y (1 + y^2)$

27 *(1st Session 2019)* If $y = \frac{10 - \cos x}{x}$, **prove that :** $x \frac{d^2 y}{d x^2} + 2 \frac{d y}{d x} = \cos x$

28 *(2nd Session 2019)* If $y = \sec x$ **Prove that :** $y \frac{d^2 y}{d x^2} + \left(\frac{d y}{d x}\right)^2 = y^2 (3 y^2 - 2)$

29 If $y = \sin 2 x$

Prove that : (1) $4 \left(\frac{d y}{d x}\right)^2 + \left(\frac{d^2 y}{d x^2}\right)^2 = 16$ (2) $\frac{d^3 y}{d x^3} + 4 \frac{d y}{d x} = 0$

30 If $y = \sin 3 x + \cos 3 x$

Prove that : (1) $\frac{d^2 y}{d x^2} + 9 y = 0$ (2) $\frac{d^4 y}{d x^4} = 81 y$

31 *(2nd Session 2008)* If $y = x \sin x$

Prove that : (1) $\frac{d^2 y}{d x^2} + y = 2 \cos x$ | (2) $x \frac{d^3 y}{d x^3} + x \frac{d y}{d x} + 2 y = 0$

(3) $\frac{d^2 y}{d x^2}$ is even function.

32 If $y = \frac{z - 1}{z}$, $z = x^2 - 1$ **Find :** $\frac{d^2 y}{d x^2}$ at $x = 0$ « 2 »

33 If $y = \sqrt{z+1}$, $z = x^2 - 10$ **Find :** $\frac{d^2y}{dx^2}$ at $x = 5$ «$\frac{-9}{64}$»

34 If $y = \tan z$, $z = 10x + 5$ **Find :** $\frac{d^2y}{dx^2}$

35 If $x = 3t^2 - 1$, $y = t^3 + 2$ **Find :** $\frac{d^2y}{dx^2}$ at $t = 4$ «$\frac{1}{48}$»

36 If $x = 2t^3 + 3$, $y = t^4$ **Find :** $\frac{d^2y}{dx^2}$ at $t = 1$ «$\frac{1}{9}$»

37 If $x = z^2 - 2z$, $y = z^2$ **Find :** $\frac{d^2y}{dx^2}$, $\frac{d^3y}{dx^3}$ at $z = 2$ «$\frac{-1}{2}$, $\frac{3}{4}$»

38 If $\frac{dz}{dx} = 1 + 2x$, $\frac{dy}{dx} = x^2 + 3$ **Find :** $\frac{d^2z}{dy^2}$ at $x = 1$ «$\frac{1}{32}$»

39 If $y = \frac{2x}{\sqrt{1+x^2}}$, $z = \frac{\sqrt{1+x^2}}{x}$ **Prove that :** $\frac{dy}{dx} = \frac{-2}{z^2} \cdot \frac{dz}{dx}$

40 If y , z and k are differentiable functions where $y = \sqrt{z}$, $z = 3 - 2k^2$, $k = 4x$

Prove that : $y\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 + 32 = 0$

41 If $x = \sec z$, $\sqrt{y} = \tan z$ **Prove that :** $\frac{d^2y}{dx^2} = 2$

42 **Choose the correct answer from the given ones :**

(1) If $f(t) = x^4$ where x is constant then $f''(t) =$ .....

(a) $12x^2$ (b) $4x^3$ (c) zero (d) $12t^2$

(2) If $y = x^2 + 2x$, and $\frac{d^2y}{dx^2} - y - 3 = 0$, then $x =$ .....

(a) 1 (b) 2 (c) -1 (d) -2

(3) If $f(x) = (x-3)^{-1}$, then $f''(3) =$ .....

(a) zero (b) -6 (c) undefined. (d) 2

(4) If $y = x^7$, $\frac{d^7y}{dx^7} =$ .....

(a) $7x^6$ (b) $42x^5$ (c) 49 (d) $\lfloor 7$

(5) If $f(x) = \cot x$, then $f''\left(\frac{\pi}{4}\right) =$ .....

(a) $\frac{-4}{9}$ (b) $\frac{4}{9}$ (c) 4 (d) $\frac{9}{2}$

(6) If $f(x) = \sin 2x \cos 2x$, then $f''\left(\frac{\pi}{3}\right) =$ .....

(a) -4 (b) zero (c) $4\sqrt{3}$ (d) 8

(7) $\frac{d}{dx}\left(\frac{dy}{dz}\right) = \frac{d^2y}{dz^2} \times$ .....

(a) $\frac{dz}{dx}$ (b) $\frac{dy}{dx}$ (c) $\frac{dx}{dz}$ (d) $\frac{dy}{dz}$

(8) If $x = (1-y)(1+y)(1+y^2)(1+y^4)$, then $\frac{d^2y}{dx^2} =$ .....

(a) $\frac{-1}{8}y^{-7}$ (b) $-56\,y^6$ (c) $\frac{-7}{64}y^{-15}$ (d) $\frac{7}{8}y^6$

(9) If $x = \sin^3\theta$, $y = \cos^3\theta$, then $\frac{d^2y}{dx^2} =$ ..... at $\theta = \frac{\pi}{4}$

(a) $\frac{2}{3}$ (b) $\frac{8}{3}$ (c) $\frac{4\sqrt{3}}{3}$ (d) $\frac{4\sqrt{2}}{3}$

(10) If $f(x) = (x-3)(x-4)(x-k)$ and $f''(3) = 2$ where $k \in \mathbb{R}$, then $k =$ .....

(a) zero (b) 1 (c) 2 (d) 3

(11) $\frac{d^2}{dx^2}(\cos^4 x + \sin^4 x) =$ .....

(a) zero (b) $-2\sin 2x$ (c) $-4\cos 4x$ (d) 1

(12) $\frac{d^3}{dx^3}(\sin^2 x) =$ .....

(a) $-4\sin x$ (b) $-4\sin 2x$ (c) $2\cos 2x$ (d) $-\sin 2x$

(13) If $x = \frac{t}{t+1}$, $y = \frac{t+1}{t}$, then $\frac{d^2y}{dx^2} = \frac{.....}{x^2}$

(a) $2x$ (b) $2y$ (c) $2x^{-3}$ (d) $-x^{-2}$

(14) If the function $f : f(x) = \begin{cases} ax^2 + bx + c & x < 0 \\ 2x + 5 & x \geq 0 \end{cases}$ is differentiable twice at $x = 0$, then $a + b - c =$ .....

(a) 7 (b) 5 (c) 3 (d) -3

(15) If $f(x)$ is a polynomial function , then $\lim_{h \to 0} \frac{f'(x+h) - f'(x)}{h} =$ .....

(a) $f'(x)$ (b) $f''(x)$ (c) $f'(h)$ (d) $f''(h)$

(16) If $f(x) = \cos 3x \cos x - \sin 3x \sin x$, then $f''\left(\frac{\pi}{4}\right) =$ .....

(a) 2 (b) 4 (c) 8 (d) 16

(17) If $y = x^n$ where $x \neq 0$, $n \in \mathbb{Z}^+$, then smallest value of n to make $\frac{d^3y}{dx^3} \neq$ zero is .....

(a) 1 (b) 2 (c) 3 (d) 4

(18) If $y = f(x)$ satisfies the relation $\frac{d^4 y}{dx^4} = y$, then y can be equal to .....

(a) $(x+1)^4$ (b) $\sin x$ (c) $\tan x$ (d) constant value $\neq$ zero

(19) If $f(x+1) = x^2 + 2x + 1$, then $f'''(3) =$ .....

(a) 1 (b) 2 (c) 3 (d) 4

(20) *(2nd Session 2024)* If $f(2x+1) = \sin 2x$, then $f'''(1) =$ .....

(a) -8 (b) -1 (c) 1 (d) 8

(21) If $0 < x < \frac{\pi}{2}$ and $f(\sin x) = \sin^2 x$, then $f''\left(\frac{1}{\sqrt{2}}\right) =$ .....

(a) 1 (b) 2 (c) π (d) $\frac{\pi}{2}$

(22) *(Trial 2021)* If $\frac{dy}{dx} = \sqrt{ay}$, $\frac{d^2 y}{dx^2} = 3$, then a = ...

(a) 6 (b) 5 (c) 3 (d) 4

(23) If $y = \frac{x-3}{x+4}$ and $(y-1)\ddot{y} = k(\dot{y})^2$, then k = ...

(a) 1 (b) 2 (c) 3 (d) 4

(24) If $3f(x) + f(4-x) = 6x^2 + 1$, then $\dot{f}(2) + f'''(2) =$ .....

(a) 3 (b) 6 (c) 12 (d) 15

(25) If $y = \tan x$, $z = \cot x$, then $\frac{d^2 y}{dz^2} =$ .....

(a) $2\tan^2 x$ (b) $-2\tan^3 x$ (c) $2\tan^3 x$ (d) $-2\cot^3 x$

(26) If f , g are two functions and $f'''(x) = -f(x)$, $\dot{f}(x) = g(x)$, then $\frac{d}{dx}\left[(f(x))^2 + (g(x))^2\right] =$ .....

(a) $\dot{f}(x) + \dot{g}(x)$ (b) 1 (c) $-f'''(x)$ (d) zero

(27) If $f(x) = 3x^4 - 2x^2 + 1$, then .....

(a) $f'''(x) = 36x^2 - 4$ (b) $f^{(4)}(x) =$ zero

(c) $f'''(0) = -4$ (d) $f^{(n)}(x) =$ zero for every $n \geq 5$

(28) If $f(x) = x^{\frac{4}{3}}$, then .....

(a) $f^{(n)}$ (zero) is undefined for every $n \geq 2$ (b) $\dot{f}$ (zero) $= \frac{4}{3}$

(c) f''' (zero) $= \frac{4}{9}$ (d) $f^{(3)}(1) = \frac{8}{27}$

(29) If $f(x) = 2\sin\frac{x}{2}\cos\frac{x}{2}$, then the 1000th derivative of this function equals .....

(a) $\sin x$ (b) $-\sin x$ (c) $-\cos x$ (d) $\sin^{-1} x$

(30) If $f(x) = \frac{x^{25}}{\lfloor 25}$, then $f^{(25)}(x) =$ .....

(a) $\lfloor 25$ (b) 25 (c) 1 (d) zero

(31) If $y = x^{n+1} + n\,x^{n-1} + 1$, then $\frac{d^n y}{d x^n} =$ .....

(a) $\lfloor n+1$ (b) $x \lfloor n+1$ (c) $x \lfloor n$ (d) $x^{-1} \lfloor n$

(32) If $y = x^n$ where n is a natural number and $\frac{d^4 y}{d x^4} = 360\, x^{n-4}$, then the value of n = ...

(a) 7 (b) 13 (c) 5 (d) 6

(33) If $f(x) = 20\, x^{n-1}$ and $\overset{\equiv}{f}(x) = c$ where $c \in \mathbb{R}$, $n \in \mathbb{Z}^+$, then n + c = ...

(a) 104 (b) 123 (c) 124 (d) 125

(34) If $y = \sin 5x$ and $\frac{d^{20} y}{d x^{20}} = a \sin 5x$, then a = ...

(a) 20 (b) 5^{20} (c) -5^{20} (d) $\lfloor 20$

(35) If (x, y) is any point on the unit circle , then .....

(a) $y\ddot{y} - 2(\dot{y})^2 + 1 = 0$ (b) $y\ddot{y} + (\dot{y})^2 + 1 = 0$

(c) $y\ddot{y} + (\dot{y})^2 - 1 = 0$ (d) $y\ddot{y} + 2(\dot{y})^2 + 1 = 0$

(36) *(1st Session 2018)* If $\dot{f}(x) = x\, f(x)$, $f(3) = -5$, then $\overset{\equiv}{f}(3) =$ .....

(a) – 50 (b) – 40 (c) 15 (d) 27

(37) If $\frac{dz}{d\theta} = \cos 2\theta$, $\frac{dy}{d\theta} = \sin 2\theta$, then $\frac{d^2 y}{d z^2} =$ ..... at $\theta = \frac{\pi}{8}$

(a) $4\sqrt{2}$ (b) 4 (c) $2\sqrt{2}$ (d) 2

(38) *(Trial 2021)* If $f\left(\frac{1}{2}x\right) = |x|^3$, then $\overset{\equiv}{f}(-1) =$ .....

(a) – 48 (b) 48 (c) 14 (d) 1

(39) If y is a function in x , then $\frac{d}{dx}\left[y^{(4)}\right] =$ .....

(a) $4\, y^{(3)}$ (b) $4\, y^{(3)} \frac{dy}{dx}$ (c) $y^{(5)}$ (d) $\frac{1}{5} y^{(5)}$

(40) If $y = x^{2021}$, then ..... = zero

(a) $y^{(2019)}$ (b) $y^{(2020)}$ (c) $y^{(2021)}$ (d) $y^{(2022)}$

(41) If $y = x^{2021}$, then ..... $= \lfloor 2021$

(a) $y^{(2019)}$ (b) $y^{(2020)}$ (c) $y^{(2021)}$ (d) $y^{(2022)}$

(42) If f , g are two differentiable functions at $x = 2$ and $h(x) = f(x) \cdot g(x)$ where $f(2) = 2\ \dot{f}(2) = 3\ \ddot{f}(2) = 12$, $g(2) = 3\ \dot{g}(2) = 4\ \ddot{g}(2) = 6$, then $\ddot{h}(2) =$ .....

(a) 66 (b) 144 (c) zero (d) 60

(43) If $x^2 + y^2 = t + \frac{1}{t}$, $x^4 + y^4 = t^2 + \frac{1}{t^2}$ where $x\,y > 0$, then $\frac{d^2 y}{dx^2} =$ .....

(a) $\frac{1}{x^3}$ (b) $\frac{2}{x^3}$ (c) $\frac{2\sqrt{2}}{x^3}$ (d) $\frac{2\sqrt{2}}{x^4}$

(44) A cube of side length ℓ length unit and its volume = V cubic unit and its total surface area = A square unit then $\frac{d^2 V}{dA^2} =$ ..... when $\ell = \frac{1}{4}$

(a) $\frac{1}{4}$ (b) $\frac{1}{12}$ (c) $\frac{1}{16}$ (d) $\frac{1}{48}$

43 *(1st Session 2008)* If $x\,y = \sin x \cos x$, **then prove that :** $x\frac{d^2 y}{dx^2} + 2\frac{dy}{dx} + 4\,x\,y = 0$

44 If $y = x \tan x$ **Prove that :** $\frac{d^2 y}{dx^2} = 2(1 + y)\sec^2 x$

45 If $x\,y = \sin a\,x$ **Prove that :** $x\frac{d^2 y}{dx^2} + 2\frac{dy}{dx} + a^2 x\,y = \text{zero}$

46 If $y = a\cos x^2 + b\sin x^2$ **Prove that :** $x\frac{d^2 y}{dx^2} - \frac{dy}{dx} + 4x^3 y = 0$

47 *(1st Session 2009)* If $x\,y = a$ (where a is (+ve) real number) , $\frac{d^2 y}{dx^2} \times \frac{d^2 x}{dy^2} > \frac{dy}{dx} \times \frac{dx}{dy}$, then find the interval to which a is belonging. « $a \in]0 , 4[$ »

48 If $y = a x^3 + b x^2$, if $\frac{d^3 y}{dx^3} = 12$, $\left(\frac{d^2 y}{dx^2}\right)_{x=1} = 6$

Find a , b « 2 , – 3 »

49 If $f(x)$ is polynomial and $f(x) + \dot{f}(x) + \ddot{f}(x) = x^3 + 7x^2 + 9x - 2$, **find** $f(x)$

50 If $y = a x^n + \frac{b}{x^{n-1}}$ **Prove that :** $x^2\frac{d^2 y}{dx^2} = n(n-1)\,y$

51 If $y = a\sin x + b\cos x$, **then prove that :** $\frac{dy}{dx} + \frac{d^3 y}{dx^3} = \frac{d^2 y}{dx^2} + \frac{d^4 y}{dx^4}$

52 If $y = f(z)$, $z = g(x)$ **Prove that :** $\frac{d^2 y}{dx^2} = \ddot{f}(z)\left[\dot{g}(x)\right]^2 + \ddot{g}(x)\,\dot{f}(z)$

53 If $y = \frac{x-1}{x+1}$ **Find :** $\frac{d^3y}{dx^3}$, $\frac{d^4y}{dx^4}$, **then deduce :** $\frac{d^n y}{dx^n}$

54 If $y = \cos ax$, discover the pattern of the successive differentiation , **then find :**

(1) $y^{(15)}$ (2) $y^{(2020)}$

55 If $y = \sin ax$ discover the pattern of the consecutive derivative **Find :**

(1) $y^{(25)}$ (2) $y^{(2016)}$

56 *(1st Session 2018)* If $\sin y + \cos 2x = 0$

Prove that : $\frac{d^2y}{dx^2} - \left(\frac{dy}{dx}\right)^2 \tan y = 4 \cos 2x \sec y$

57 If $y = \tan^2 x$ **Prove that :** $\frac{1}{2}\frac{d^2y}{dx^2} = (1 + y)(1 + 3y)$

Higher skills

58 **Choose the correct answer from the given ones :**

(1) If $y = 1 + \frac{x}{\lfloor 1} + \frac{x^2}{\lfloor 2} + \frac{x^3}{\lfloor 3} + \cdots$ to ∞

, then $5\ddot{y} - 2\dot{y} - 3y =$ .....

(a) zero (b) 2 y (c) 6 y + 3 (d) 2 y – 1

(2) If $y = f(x)$ and $f(x + h) - f(x) = 5x^2 h + h^2$, then $\frac{d^3y}{dx^3} =$ .....

(a) 6 (b) 10 (c) 14 (d) 18

(3) If f , g are two functions where $f(x) = x^2$, $g(2) = 3$

, $\dot{g}(2) = -2$, $\ddot{g}(2) = 5$, then $(f \circ \ddot{g})(2) =$ .....

(a) 16 (b) 24 (c) 32 (d) 38

(4) If $\lim_{h \to 0} \frac{\dot{f}(1) - \dot{f}(1+h)}{2h} = 21$ where $f(x) = 2x^4 - ax^3$, then a = ...

(a) 11 (b) 12 (c) 13 (d) 14

(5) If $\sqrt{y + x} + \sqrt{y - x} = k$, then $\frac{d^2y}{dx^2} =$ .....

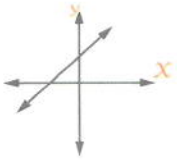
(a) k^2 (b) $\frac{1}{2}k^2$ (c) $\frac{2}{k^2}$ (d) $\frac{1}{k^2}$

(6) If $y = \sin 2x$ and $\frac{d^n y}{dx^n} = 2^n \sin 2x$, then .....

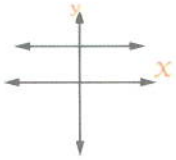
(a) n is an even number. (b) n is an odd number.

(c) n is divisible by 3 (d) n is divisible by 4

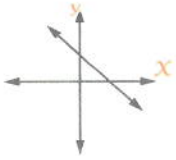
(7) If $y = a x^n - b x^{n+1} + 5$ is a polynomial and $a , b \in \mathbb{R}^+$, then $\frac{d^n y}{d x^n}$ may by represented by the figure .....



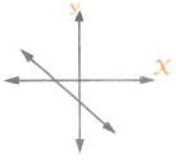
(a)



(b)



(c)



(d)

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59 If $y = \left(x + \sqrt{1 + x^2}\right)^5$ **Prove that :** $(1 + x^2)\frac{d^2 y}{d x^2} + x\frac{d y}{d x} - 25\, y = 0$

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60 If $x = \sin y$, y is an acute angle. **Prove that :** $\frac{d^2 y}{d x^2} = \frac{x}{(1 - x^2)^{\frac{3}{2}}}$

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61 *(1st Session 2016)* If $y = \frac{a x + b}{c x + d}$ where a , b , c , d are constants

Prove that : $2\left(\frac{d y}{d x}\right)\left(\frac{d^3 y}{d x^3}\right) = 3\left(\frac{d^2 y}{d x^2}\right)^2$